
GEOG 311 Final Project Paper

Student Housing Optimization: Multi-Criteria Spatial Analysis of Student Housing Factors

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Article Information <i>Article History:</i> Received 22 April 2025 Keywords: Student Housing, Multi-Criteria Decision Analysis, Spatial Analysis, Weighted Overlay	Abstract Searching for suitable housing poses a challenge for many university students due to a wide variety of both spatial and personal constraints. The model created in this project offers a GIS-based weighted multi-criteria decision analysis (MCDA) model designed to optimize the selection of student housing surrounding Brigham Young University (BYU). The model functions by incorporating seven key factors: proximity to campus (35%), housing cost (25%), transportation access (15%), commercial amenities (10%), recreation facilities (10%), proximity to churches (5%), and parking availability (5%). Implementation of GIS analysis techniques such as network analysis, near table generation, and various spatial joins allow for the identification of optimal housing zones within the study area. Results show that areas within 800m of BYU campus prove to be optimal locations for student housing, receiving the highest suitability scores (8.5-9.8). Deeper analysis of individual factors supports the idea that these optimal housing zones offer a balance of accessibility and overall location quality. This research helps to contribute to our understanding of the spatial factors that influence student housing preferences and academic success through its successful implementation of multi-criteria decision analysis. Limitations of the project include data availability, field computation, and the subjectivity of weighting various factors. These limitations suggest opportunities for further refinement and improvement of this model.
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1. Introduction

Students spend countless hours each semester searching for the perfect place to live. With each student having different needs and requirements for student housing, the current methods of locating student housing are ineffective and outdated. This model utilizes a multi-criteria weighted model to effectively locate ideal housing based on parameters such as cost, amenities, room type, and other important factors.

1.1 Previous Research

In a research study done by consulting company InnesWorks, it was determined that the location which students are living can have an impact on the likelihood of students to skip class and reduce their likelihood of engaging in school activities with students located further from campus being less likely to attend their classes.

Shimberg Florida created a similar Housing Suitability Model (HSM) created for Housing Studies utilizing GIS analysis to analyze factors like transit accessibility, driving costs, and neighborhood socioeconomic characteristics, income levels, crime risk, and poverty

to assess the suitability of new housing locations. It concludes that Central city sites are higher in transit accessibility and lower driving costs but lack socioeconomic characteristics.

1.2 Summary and Relevance of Previous Research

Studies show that student housing location has a direct impact on student performance in school. Similar HSM models have been used for different purposes, mainly for assessing the areas to build new sites. Most of the models utilize common factors to assess an area's suitability but lack the specification needed for BYU student housing.

Student Housing Factors

Factor (Weight)	Calculation Method
Proximity to BYU (35%)	Distance from Library: $\leq 500\text{m} = 3.5\text{pts}$ $\leq 800\text{m} = 2.8\text{pts}$ $\leq 1500\text{m} = 2.1\text{pts}$ $> 1500\text{m} = 1.4\text{pts}$
Housing Cost (25%)	$\leq \$332 = 2.5\text{pt}$ $\$332-775 = \text{Linear scale}$ $\geq \$775 = 0.5\text{pts}$
Transportation (15%)	Bus stops within 500m (normalized to 1.5pts max)
Commercial (10%)	Commercial locations within 800m (normalized to 1.0pts max)
Recreation (10%)	Recreation facilities within 800m (normalized to 1.0pts max)
Churches Nearby (5%)	Churches within 500m (normalized to 0.5pts max)
Parking Nearby (5%)	Parking areas within 500m (normalized to 0.5pts max)

Table 1. Student housing factors and associated weights.

2. Conceptual Model

2.1 Study Area:

The study area for this model encompasses the Provo and Orem area, cities that serve as the primary residential zones for BYU students. This area covers

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approximately 72 square miles and features a few unique urban development patterns.

2.1 a.) BYU Campus Central Zone: Area that is densely populated with students and student housing, the area immediately surrounding BYU campus.

2.1 b.) Residential Transition Zone: Approximately 1-3 miles of BYU, comprising a mix of student and family housing, townhomes, and single family homes.

2.1 c.) Commercial Zone: Large retail and transportation hubs, areas for recreation and gathering nearby central areas.

2.2 Spatial Resolution

This model uses 30-meter grids for its base suitability model, this allows for neighborhood-level differentiation. It also utilizes point data for identifying specific housing complexes and points of interest relevant to the model. Additionally, the model employs a system of network-based distance calculations to find accurate measurements of proximity.

2.3 Data Precision

Most of the factors will have spatial precision at a building-level with some variance for factors such as UTA Transit Routes being slightly less precise as well as crime statistics which are on a block group level.

2.4 Modeling Approach

The model utilizes a weighted criteria model within a multi-criteria decision analysis model in order to carefully weigh each independent factor prior to computing the suitability of a location. All of the factors will be converted to a 10-point suitability scale with one decimal point.

2.5 Validation

In order to validate results, the scores assigned by the model will be compared to ratings of housing complexes and student surveys.

2.6 Data Sources:

BYU Contracted Housing Database (2024): contains location, cost, and type information for all BYU-approved housing complexes.

Utah Transit Authority (UTA) GTFS data (2024): current bus routes and stop locations.

Utah Automated Geographic Reference Center (AGRC) datasets (2024): commercial zones, recreation facilities, and parking areas.

OpenStreetMap (accessed March 2025): additional point-of-interest data for churches and amenities.

Provo City GIS Department (2024): road network data for network analysis.

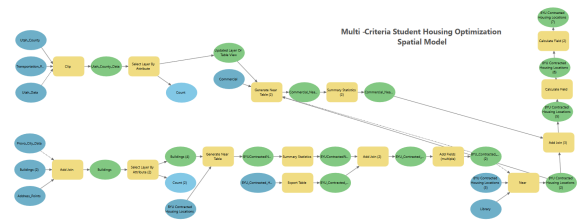


Fig. 2 Work-flow representing the GIS-based model.

3.1 Mathematical Model

The suitability score (S) for each housing location:

$$S = \sum_{i=1}^7 W_i \cdot F_i$$

Where:

S = Total suitability score W_i = Weight of factor i
 F_i = Score for factor i i = Factor index (1-7)

Eq. 1. Equation used to calculate suitability scores within the model.

3. GIS-Based Modeling Approach

The multi-criteria student housing optimization model utilizes a weighted overlay approach to account for various spatial datasets. By using this approach, geoprocessing tools can be used on individual datasets giving deeper analysis of each layer. After geoprocessing individual datasets, the resulting tables are joined to a main table where field calculation occurs. This approach also allows for easily modifying factors and their corresponding weights, ultimately allowing for greater application and usage. The workflow for this model (Fig. 2) is simple and allows for easy changing of datasets, weights, and locations.

3.2 GIS Implementation:

GIS analysis techniques used in this model:

Python Scripting: various Python scripts were utilized to calculate scores for factor analysis and combining datasets.

Near Table Analysis: generation of near tables allowed for deeper analysis and statistics regarding each contracted housing location.

Buffer: utilized the buffer tool to create specific study areas and analyze various factors.

Network Analysis: approximation of travel times and distances were calculated using network analysis geoprocessing tools.

Spatial Join: multiple spatial joins and queries were required to combine datasets and increase computational efficiency.

Clip: the clip geoprocessing tool allowed for clipping layers to meet the predefined study area.

Attribute Queries: in addition to spatial queries, attribute queries were utilized to help improve specificity and precision in analysis.

4. Results

The model yielded a comprehensive spatial suitability assessment (Fig. 3) of the BYU contracted student housing facilities in the Provo area. Each complex was evaluated using a 10-point suitability scale with one decimal point based on weighted criteria of accessibility, housing characteristics, and overall location quality factors.

4.1 Spatial Analysis of Housing Suitability

Spatial analysis underscored distinct patterns regarding overall housing suitability across the study area. Complexes that scored highly (8.5-9.8) appear in a cluster immediately South of BYU campus. The highest score of 9.8 was Campus Plaza Apartments. This result aligns with the expected results as it is the closest location geographically and has the cheapest average cost.

Outside of this cluster, the suitability of housing decreases (8.0-8.4) in a fan-like pattern as it extends further South and East of BYU campus. This also aligns with expected results as scores decrease as distance from campus increases.

Complexes that scored the lowest (4.7-6.6) tend to be located around the edges of the study area, particularly further East or West of BYU campus.

Housing complexes with high suitability scores located further from BYU campus can be found but their scores can be attributed to higher scores in specific categories such as increased commercial availability or low average costs.

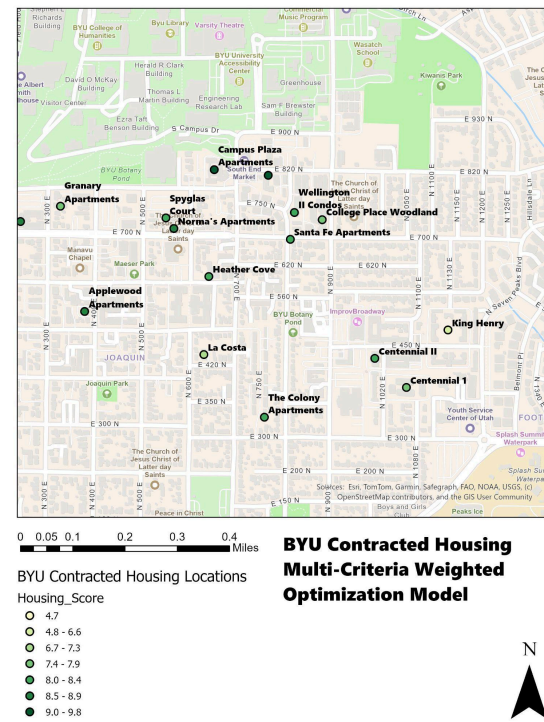


Fig. 3. Visual of suitability assessment for the Provo area.

4.2 Factor Analysis:

Analysis of individual factors towards the overall suitability score showed the following patterns:

Accessibility Factors:

Complexes that scored highly (above 8.0) were consistently located close to campus with ample transportation options. Travel times to the center of BYU for these complexes were on average less than 10 minutes. Transportation, another accessibility factor, was moderately correlated with overall suitability as complexes located near popular UTA routes scored higher than complexes located further away.

Housing Characteristics:

The most important factor in terms of housing characteristics cost aligned with expected results as the highest scoring complexes were not only the lowest cost but also offered amenities such as laundry.

Location Quality:

Overall location quality appears to be consistent as well with higher scoring complexes located closer to major commercial areas, parking, recreation, and with multiple churches in the surrounding area.

5. Discussion

The model developed in *Student Housing Optimization: Multi-Criteria Spatial Analysis of Student Housing Factors* developed in this study helps to give valuable insight into the ideal locations and most important factors relating to BYU-contracted student housing in Provo. Results validate methodology of the underlying model as the emerging spatial patterns closely follow expectations regarding proximity, accessibility, and overall location quality.

5.1 Interpretation of Results

Spatial patterns demonstrated by this model closely align with previous studies of similar topics in student housing. High suitability scores within 800m of BYU campus support previous findings that location impacts overall student engagement. In addition to that, the correlation between accessibility and suitability also follows previous findings of transit accessibility.

The model demonstrates that the factors housing cost and proximity are closely related as housing complexes located in closer proximity to campus have higher average prices than those located further away. The same complexes received a higher overall suitability score, demonstrating the importance of proximity to campus in overall student housing suitability.

5.2 Limitations:

In the process of implementing the model, various challenges became apparent:

Data Engineering:

Finding quality and thorough data of housing complexes proved to be challenging with most datasets containing only partial data. Joining the data also proved to be challenging due to each dataset using different formats and methods for storage. Copious amounts of data engineering and formatting were required to create a comprehensive and usable dataset.

Weighting of Factors:

Establishing a set weight of each factor was difficult due to the subjective nature of the topic. This challenge was overcome by incorporating weights based on prior research studies and popular opinions on the matter. This subjectivity can also be overcome through changing weights to meet more specific use cases of the model.

Field Calculation:

Challenges regarding computing the overall suitability scores were apparent as computation limits were imposed by ArcGIS Pro dialogs. This resulted in tedious manual computation and heavy reliance on Python scripting. This can be mitigated in later iterations through more effective equation design and better data storage practice.

Despite these challenges, the model aligned well with expected behavior and successfully identified various patterns regarding student housing suitability at BYU.

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